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Editor-in-Chief
Information Processing & Management

Dear Editor,

Please consider the enclosed manuscript, “Causal graph recommendation under exposure bias: removing the repeated-walk bias of doubly robust propagation”, for publication in *Information Processing & Management*.

Graph-based recommenders propagate preferences over interaction graphs that the platform’s own exposure policy has filtered. The manuscript shows that the inverse-propensity and doubly robust corrections used inside graph propagation are themselves biased once propagation spans more than one hop, gives the exact bias, and removes it with a training-free operator (DRUP) that is exact for any odd number of hops and, at three hops, costs no more than the uncorrected operator. It states the guarantees that follow, with their assumptions: edge-wise double robustness, robustness to within-user exposure dependence, and exposure invariance in expectation, plus the properties that any linear operator with fixed nuisances shares (exact attributions, certificates against injected profiles, joint differential privacy with public nuisances, capped allocation). It also shows, by simulation, that these guarantees require nuisances fitted independently of the log: they hold for an independent log or a sample split, but not for nuisances cross-fitted on the same log, and a sample-split variant is evaluated alongside.

The evaluation is deliberately conservative. A semi-synthetic test on KuaiRec with known exposure shows that the correction removes the 18–29% overestimate of the full-exposure utility of fixed allocations that uncorrected operators produce. On Coat, Yahoo! R3, KuaiRand and KuaiRec, whose test data are random or fully observed, DRUP is compared with up to 26 tuned baselines under one protocol (per-split tuning and early stopping of every method, user-level tests with Holm correction over all comparisons). The manuscript reports that the walk correction does not change top- K accuracy or rankings, so that its contribution is at the level of scores; that the most accurate method differs by dataset (iALS on Coat, BPR-trained graph models on Yahoo! R3 and KuaiRand, training-free propagation over the doubly robust graph on KuaiRec); that the sample split that the guarantees require costs accuracy on sparse logs; and that a real-data thinning audit of exposure invariance has limited power, and the selected configurations are not invariant where they use clipped inverse-propensity edges.

We believe the work fits the journal’s interest in trustworthy information access and recommender systems. The manuscript is original, has not been published elsewhere and is not under consideration by another journal. Code and results are publicly available. In line with the double-anonymised review policy, the manuscript file contains no identifying information; author details and declara-

tions are on the separate title page.

Sincerely,

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